ERD & Schema Design

WatchShop — Watch Business Data Management System

Group 8

CSE311L.3

Md Roman (ID: 2312765042)

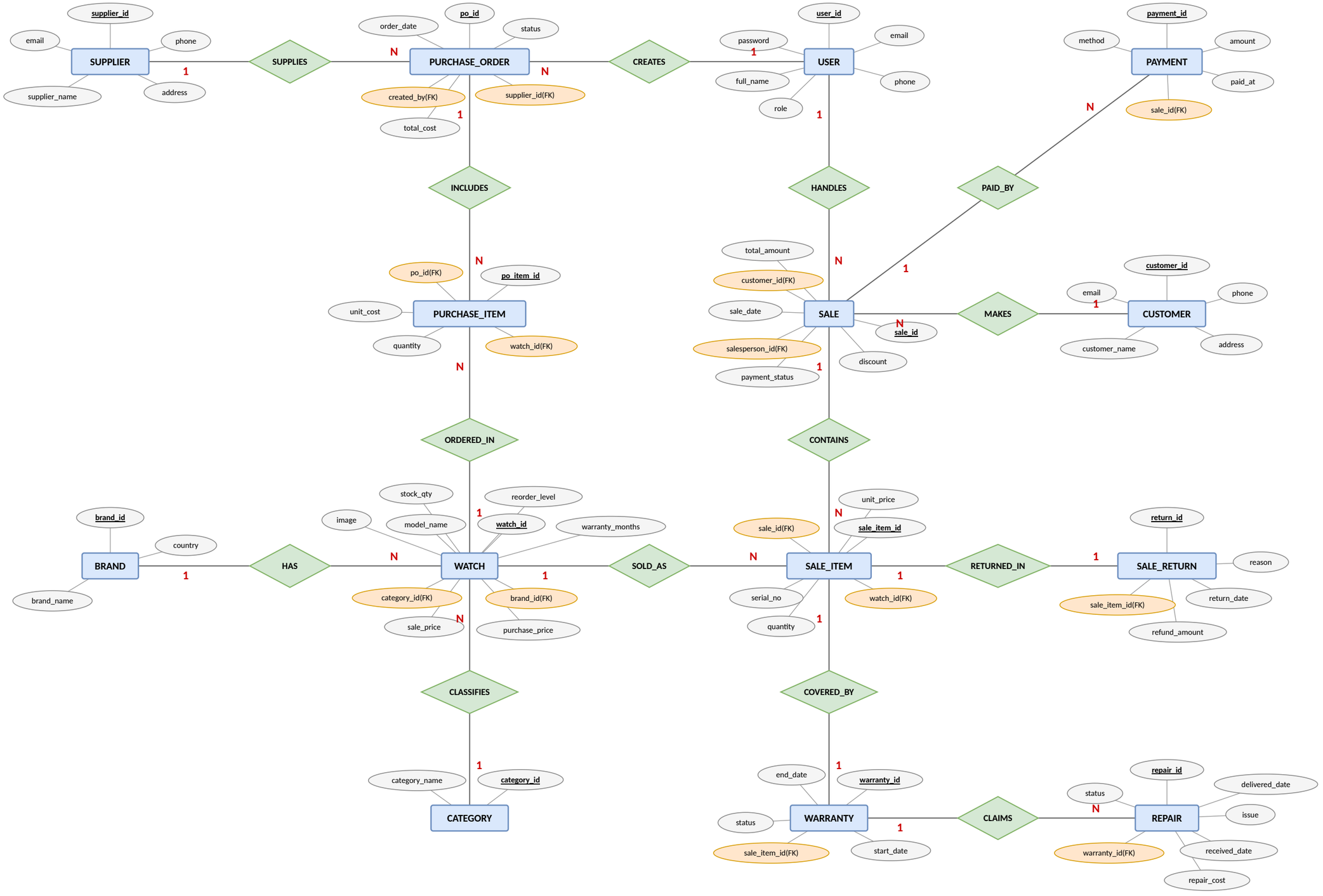
Anika Nower Nihan (ID: 2011234642)

Farhan Shahriar (ID: 2312071042)

Ashiqur Rahman (ID: 2412388042)

Faculty: Ms. Tanzilah Noor Shabnam [TnS1] | Lab Instructor: Md. Amanullah Shah
North South University · Submission: November 5, 2026

ER - Diagram of 'WatchShop' (Group-8)



Notation (Chen): Rectangle = Entity, Diamond = Relationship, Ellipse = Attribute, Underlined = Primary Key, Orange ellipse = Foreign Key, 1 / N = Cardinality.

1. Entities & Attributes

Entity	Primary Key	Attributes
SUPPLIER	supplier_id	supplier_name, phone, email, address
PURCHASE_ORDER	po_id	supplier_id, created_by, order_date, status, total_cost
USER	user_id	full_name, email, password, phone, role
PAYMENT	payment_id	sale_id, amount, method, paid_at
PURCHASE_ITEM	po_item_id	po_id, watch_id, quantity, unit_cost
SALE	sale_id	customer_id, salesperson_id, sale_date, discount, total_amount, payment_status
CUSTOMER	customer_id	customer_name, phone, email, address
BRAND	brand_id	brand_name, country
WATCH	watch_id	model_name, brand_id, category_id, purchase_price, sale_price, stock_qty, reorder_level, warranty_months, image
SALE_ITEM	sale_item_id	sale_id, watch_id, serial_no, quantity, unit_price
SALE_RETURN	return_id	sale_item_id, return_date, reason, refund_amount
CATEGORY	category_id	category_name
WARRANTY	warranty_id	sale_item_id, start_date, end_date, status
REPAIR	repair_id	warranty_id, received_date, issue, repair_cost, status, delivered_date

2. Relationships & Cardinality

Entity A	Relationship	Entity B	Cardinality	Meaning
SUPPLIER	SUPPLIES	PURCHASE_ORDER	**1 : N**	One supplier supplies many purchase orders.
USER	CREATES	PURCHASE_ORDER	**1 : N**	One admin creates many purchase orders.
PURCHASE_ORDER	INCLUDES	PURCHASE_ITEM	**1 : N**	One purchase order includes many line items.
WATCH	ORDERED_IN	PURCHASE_ITEM	**1 : N**	One watch model appears in many purchase line items.
BRAND	HAS	WATCH	**1 : N**	One brand has many watch models.
CATEGORY	CLASSIFIES	WATCH	**1 : N**	One category classifies many watches.
USER	HANDLES	SALE	**1 : N**	One salesperson handles many sales.
CUSTOMER	MAKES	SALE	**1 : N**	One customer makes many sales.
SALE	PAID_BY	PAYMENT	**1 : N**	One sale can be paid in many (partial) payments.
SALE	CONTAINS	SALE_ITEM	**1 : N**	One sale invoice contains many items.
WATCH	SOLD_AS	SALE_ITEM	**1 : N**	One watch model is sold in many sale items.

SALE_ITEM	RETURNED_IN	SALE_RETURN	**1 : 1**	A sold item can be returned at most once.
SALE_ITEM	COVERED_BY	WARRANTY	**1 : 1**	Each sold item gets exactly one warranty.
WARRANTY	CLAIMS	REPAIR	**1 : N**	One warranty can have many repair requests.

Many-to-many: PURCHASE_ORDER ↔ WATCH (M:N) is resolved by PURCHASE_ITEM, and SALE ↔ WATCH (M:N) is resolved by SALE_ITEM.

3. Foreign Keys

Table	Foreign Key	References
PURCHASE_ORDER	supplier_id	SUPPLIER(supplier_id)
PURCHASE_ORDER	created_by	USER(user_id)
PAYMENT	sale_id	SALE(sale_id)
PURCHASE_ITEM	po_id	PURCHASE_ORDER(po_id)
PURCHASE_ITEM	watch_id	WATCH(watch_id)
SALE	customer_id	CUSTOMER(customer_id)
SALE	salesperson_id	USER(user_id)
WATCH	brand_id	BRAND(brand_id)
WATCH	category_id	CATEGORY(category_id)
SALE_ITEM	sale_id	SALE(sale_id)
SALE_ITEM	watch_id	WATCH(watch_id)
SALE_RETURN	sale_item_id	SALE_ITEM(sale_item_id)
WARRANTY	sale_item_id	SALE_ITEM(sale_item_id)
REPAIR	warranty_id	WARRANTY(warranty_id)

4. Logical Schema - WatchShop

Database Schema :

SUPPLIER (**supplier_id** [PK], supplier_name, phone, email, address)

PURCHASE_ORDER (**po_id** [PK], supplier_id [FK → SUPPLIER.supplier_id], created_by [FK → USER.user_id], order_date, status, total_cost)

USER (**user_id** [PK], full_name, email, password, phone, role)

PAYMENT (**payment_id** [PK], sale_id [FK → SALE.sale_id], amount, method, paid_at)

PURCHASE_ITEM (**po_item_id** [PK], po_id [FK → PURCHASE_ORDER.po_id], watch_id [FK → WATCH.watch_id], quantity, unit_cost)

SALE (**sale_id** [PK], customer_id [FK → CUSTOMER.customer_id], salesperson_id [FK → USER.user_id], sale_date, discount, total_amount, payment_status)

CUSTOMER (**customer_id** [PK], customer_name, phone, email, address)

BRAND (**brand_id** [PK], brand_name, country)

WATCH (**watch_id** [PK], model_name, brand_id [FK → BRAND.brand_id], category_id [FK → CATEGORY.category_id], purchase_price, sale_price, stock_qty, reorder_level, warranty_months, image)

SALE_ITEM (**sale_item_id** [PK], sale_id [FK → SALE.sale_id], watch_id [FK → WATCH.watch_id], serial_no, quantity, unit_price)

SALE_RETURN (**return_id** [PK], sale_item_id [FK → SALE_ITEM.sale_item_id], return_date, reason, refund_amount)

CATEGORY (**category_id** [PK], category_name)

WARRANTY (**warranty_id** [PK], sale_item_id [FK → SALE_ITEM.sale_item_id], start_date, end_date, status)

REPAIR (**repair_id** [PK], warranty_id [FK → WARRANTY.warranty_id], received_date, issue, repair_cost, status, delivered_date)